

→ Logistic Regression
It is supervised classification model.

• dependent variable is categorical and binary (0 or 1)

it is used when your output is

- Yes/no
- 0/1
- True/false
- Pass/fail.

ex:

- will student pass? (Yes/no)
- Email spam hai ya nahi.

• instead of predicting a number like.

marks = 75

Logistic Regression predicts
probability = 0.85 (85% chances of Yes)
Then Decision

- if probability $\geq 0.5 \rightarrow$ Yes (1)
- if probability $< 0.5 \rightarrow$ No (0).

→ Why not linear Regression?
 Agar hum linear Regression use karenge

$$y = \alpha x + 1$$

output could be

- -10 x

- 150 x

But probability sirf 0 se 1 ke beech honi chaye.

Is problem ko solve karne hai logistic Regression.

Solution!

Sigmoid function (Magic step)
Logistic Regression ek special function
use krta hai.

$$\sigma(Z) = \frac{1}{1 + e^{-Z}}$$

where

- $Z = mx + b$. (same Linear Regression)
- output always b/w 0 and 1

Simple flow:

1. pehle Linear Regression
krate hai!

$$Z = mx + b$$

2. phir us ko sigmoid mai
convert krta hai.
daalte

$$\text{probability} = \text{sigmoid}(Z)$$

→ Find decision:

probability:

probability $\geq 0.5 \rightarrow 1$

probability $< 0.5 \rightarrow 0$

→ Example of Logistic Regression

Result.

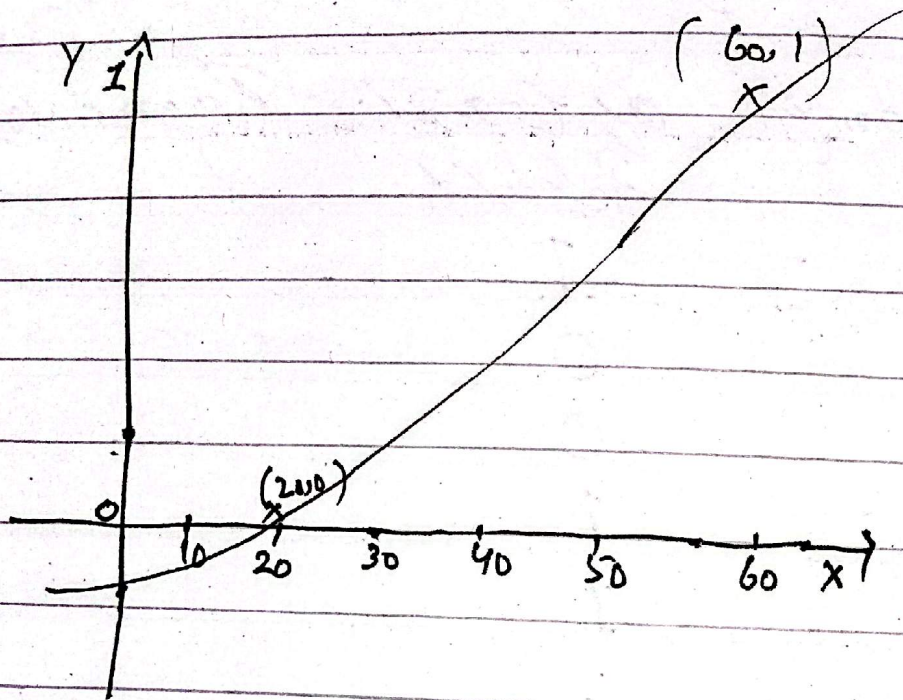
Age	Result
50	1
30	0
20	0
40	1
60	1

→ Now if we will ask to model after training if there is a person whose marks is 15, does he will come into the pass category or fail.
So this is what we have to find.

Formula: $Z = \frac{1}{1 + e^{-n}}$

$Z = \underbrace{b_0}_{-0.5} + \underbrace{b_1}_{0.25} \underbrace{X_1}_{\text{Age (15)}}$

input variable: X_1



Now we can easily draw the slope of that line!

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{1 - 0}{60 - 20} = \frac{1}{40} = \boxed{0.025}$$

$$y - y_1 = m(x - x_1)$$

$$y - 0 = 0.025(x - 20)$$

$$y = 0.025x - 0.5$$

solution:

$$f(n) = \frac{1}{1 + e^{-[-0.5 + (0.025)(15)]}}$$

$$= \frac{1}{1 + e^{-(-0.125)}}$$

$$= \frac{1}{1 + 1.1331} = \frac{1}{2.1331} = \boxed{0.468}$$

it is like $0.5 <$

so is the interpretation $\log$ Fail.

so if $\frac{1}{1 + e^{-[-0.5 + (0.025)(45)]}}$

$$= \frac{1}{1 + e^{-0.625}} = \frac{1}{1.5352} = 0.65$$

$\boxed{\text{Pass}}$
belongs to 1.